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Predicting length of stay and mortality among hospitalized patients with type 2 diabetes mellitus and hypertension

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ABSTRACT

Background: Type 2 diabetes mellitus (T2DM) and hypertension (HTN), both non-communicable diseases, are leading causes of death globally, with more imbalances in lower middle-income countries. Furthermore, poor treatment and management are known to lead to intensified healthcare utilization and increased medical care costs and impose a significant societal burden, in these countries, including Indonesia. Predicting future clinical outcomes can determine the line of treatment and value of healthcare costs, while ensuring effective patient care. In this paper, we present the prediction of length of stay (LoS) and mortality among hospitalized patients at a tertiary referral hospital in Tasikmalaya, Indonesia, between 2016 and 2019. We also aimed to determine how socio-demographic characteristics, and T2DM- or HTN-related comorbidities affect inpatient LoS and mortality.
Methods: We analyzed insurance claims data of 4376 patients with T2DM or HTN hospitalized in the referral hospital. We used four prediction models based on machine-learning algorithms for LoS prediction, in relation to disease severity, physician-in-charge, room type, co-morbidities, and types of procedures performed. We used five classifiers based on multilayer perceptron (MLP) to predict inpatient mortality and compared them according to training time, testing time, and Area under Receiver Operative Curve (AUROC). Classifier accuracy measures, which included positive predictive value (PPV), negative predictive value (NPV), F-Measure, and recall, were used as performance evaluation methods.
Results: A Random forest best predicted inpatient LoS (R2, 0.70; root mean square error [RMSE], 1.96; mean absolute error [MAE], 0.935), and the gradient boosting regression model also performed similarly (R2, 0.69; RMSE, 1.96; MAE, 0.935). For inpatient mortality, best results were observed using MLP with back propagation (AUROC 0.899; 69.33 and 98.61 for PPV and NPV, respectively). The other classifiers, stochastic gradient descent with regression loss function, Huber, and random forest models all showed an average performance.
Conclusions: Linear regression model best predicted LoS and mortality was best predicted using MLP. Patients with primary diseases such as T2DM or HTN may have comorbidities that can prolong inpatient LoS. Physicians play an important role in disseminating health related information. These predictions could assist in the development of health policies and strategies that reduce disease burden in resource-limited settings.

Abbreviations: **CKD**, chronic kidney disease; **COPD**, chronic obstructive pulmonary disease; **CVD**, cardiovascular disease; **HTN**, hypertension; **ICD**, International Classification of Diseases; INA-CBGs, Indonesia Case Base Groups; LoS, length of stay; **MLP**, multilayer perceptron; **NCD**, non-communicable; **RSUD**, rumah sakit umum daerah; **SVM**, support vector machine; **T2DM**, type 2 diabetes mellitus.

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1. Introduction

Non-communicable diseases (NCDs) have been reported to be leading causes of death globally, with more deaths per year than all other death-related causes combined [1]. An increasing and unbalanced effect of NCDs is reported to specifically affect lower middle-income countries (LMICs), of which 75% involve cardiovascular- and diabetes-related mortalities [2]. In Southeast Asia, an increase in NCDs has been observed, and only limited resources have been utilized to address this rising NCD pandemic and associated health imbalances [3]. In Indonesia, as in other LMICs, social determinants of health (such as socioeconomic factors, education, neighborhood, employment, physical environment, social networks, etc.) and nationwide health imbalances (shortage of health workforce as compared to the actual demand) have not been fully addressed at national level. This has made developing and implementing policies to decrease health disparities challenging in relation to NCDs and their negative effects on wellbeing and on Indonesia's financial structure [4].

NCDs are linked to heavy economic burdens on households [5]. Costs incurred following hospitalization have been reported to be significantly higher for patients with NCDs than for those with communicable diseases (CDs). Compared to that for inpatients with a CD, a single hospital stay for an inpatient with an NCD has been shown to result in almost twice the out-of-pocket (OOP) expenses [6]. Government social insurance schemes are very limited and are often not utilized. Consequently, 8% of Indonesia's OOP healthcare expenditure and 47.3% within various households has been directed towards NCD management [6,7]. It has been estimated that this current 8% OOP expenditure is likely to have increased to 12% by 2020 without a significant health intervention [7]. Type 2 diabetes mellitus (T2DM) and hypertension (HTN) are chronic diseases that require ongoing clinical observation and patient self-management [8], and both place a considerable economic burden on medical service frameworks. The direct costs of these chronic NCDs to the patient, the family, and the government include hospital and medication costs. Many employed patients of working age may be absent from work or have reduced productivity due to illness or disability. Therefore, estimating the economic burden due to hospital admissions is important in the current setting through predicting inpatient length of stay (LoS) and mortality, which has been under-researched in developing countries [9].

Indonesia continues to confront a rapidly growing and costly increase in health expenditure due to cardiovascular disease (CVD), T2DM, and other NCDs [10,11]. Since 2006, there has been a 38.5% increase in DM-related complications and fatalities. Moreover, as compared to the Indonesia Basic health Research reports in 2013, those in 2018, showed growing trends for most NCDs such as cancer, stroke, kidney disease, joint diseases, DM, CVD, HTN and obesity [10,12], which is likely to place an additional and long-term economic burden on Indonesia's healthcare system. It has been reported that, of 230 million National Health Insurance participants, 11 million people have been diagnosed with HTN and 9 million people have been diagnosed with T2DM [13].

Despite HTN being a preventable disease, recent studies have reported low prevalence rates for antihypertensive medication treatment [14]. The prevention and treatment of T2DM and HTN are limited due to inadequate financing of pharmacotherapies, and also because in LMICs such as Indonesia, it has been reported that most healthcare workers are not sufficiently well trained to detect cases and treat patients [15]. Poor treatment and management of T2DM and HTN are known to lead to intensified healthcare utilization and increased medical care costs and impose a significant societal burden [16]. Therefore, predicting future clinical outcomes form an important aspect in determining the line of treatment and value of healthcare costs.

Various risk factors have been associated with HTN and T2DM, including socio-demographic risk factors [17,18], stress [19] and obesity [14], dietary behaviors [14], physical inactivity and cigarette smoking [20]. The presence of one risk factor increases the risk of developing other complications. This close relationship between T2DM and HTN suggests a possible common genetic or pathophysiological process or both. HTN and T2DM are associated with an increased risk of CVD and renal disease, and the risks are exacerbated when both complications are present [21]. Due to the close relationship between these two diseases, a combined analysis could be useful to predict the further outcomes.

Patient age, clinical and biochemical features, co-morbidity of T2DM and HTN, presence of other co-morbidities, and type of treatment have been found to significantly affect inpatient LoS and mortality [22–25]. T2DM and HTN incur medical costs not only because of direct treatment but also because of treatment of sequelae such as CVD, cerebrovascular (CVA) disease, and renal disease, which are complications that occur commonly in patients with T2DM and HTN [26]. Chronic complications require costly medical procedures, such as limb amputation or renal dialysis, and significantly increase hospitalization costs, inpatient LoS, and mortality [27–30]. This burden of indirect costs causes a loss of productivity due to disability and premature mortality [31]. Therefore, determining inpatient LoS and mortality is likely to help in terms of prevention and management of complications for patients with T2DM and/or HTN.

Deep learning models have been used to predict mortality, LoS, and 20 ICD-9 codes group by using Medical Information Mart for intensive Care III (MIMIC-III) dataset [32].. Studies also have analyzed factors influencing inpatient LoS [14,17–19,33,34], the complications and healing for patients with both T2DM and HTN [21]. In Indonesia, the universal health care system, namely, the Indonesia Case Base Group (INA-CBGs) dataset, was developed with the participation of health professional organizations and hospital associations and has been widely used for purpose of conducting research studies. [35,36]. Using this dataset, a study examined pharmaco-economic aspects of analyzing the cost-effectiveness of pharmacist counseling services in terms of the success of therapy and the quality of life for patients with HTN [37].

We further reviewed the literature to identify studies that had used data from referral hospitals in Indonesia. To date, no previous studies based on Indonesian datasets have reported predictions concerning inpatient LoS or mortality for patients with T2DM or HTN. These predictions are essential to assist in the development of health policies and strategies that reduce disease burden in resource-limited settings.

This study aimed to predict inpatient LoS and mortality among patients admitted to a tertiary referral hospital in Tasikmalaya (West Java) between 2016 and 2019. We also aimed to describe how socio-demographic characteristics and comorbidities associated with patients with T2DM or HTN as a primary diagnosis affect inpatient LoS and mortality. Finally, we aimed to analyze patients with T2DM and/or HTN to determine the combined effect of these diseases.

2. Methods

2.1. Indonesian case base groups (INA-CBGs) claims dataset

The source of our insurance claims data was the referral hospital in Indonesia- Dr Soekardjo Regional Public Hospital- (RSUD Dr. Soekardjo), which is a state-owned type B regional public hospital with 453 beds and located in the Tasikmalaya area [38].

Data included admission and discharge dates, birthdates, age, sex, diagnosis codes or the International Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM), procedure codes or International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM), type of hospital bed, INA-CBGs code (containing the code for the primary diagnosis) with a combination of a letter and number along with information concerning the patient type (namely, procedure inpatient, non-procedure inpatient, midwifery inpatient, or neonatal inpatient), severity level, discharge status (physician's permission to discharge/instructions, referral, patient request, death, others), medical

record number, and the physician-in-charge on duty (the physician treating the patient). All collected data were anonymized.

2.2. Study design and population

A cohort of hospitalized patients was retrospectively selected from the insurance claims data of the RSUD Dr. Soekardjo hospital that provided extensive specialist medical services and limited subspecialist services between January 1, 2016 and December 31, 2019, the time period for which data was available. Inclusion criteria were as follows: (i) a primary discharge diagnosis of T2DM (E11) or (ii) a primary diagnosis of HTN (I10-I15), according to ICD-10-CM codes. The study sample was derived from a parent cohort that has been used to develop and validate artificial intelligence models, which was not restricted to patients with a primary discharge diagnosis of T2DM and or HTN, and the methods employed to create and analyze subsets were similar to those of the parent cohort. Index discharge for patients with a discharge status of 4, according to INA-CBGs codes that signified death on discharge, were excluded for prediction of inpatient LoS, as were discharge due to transfer to another hospital with a discharge status of 2 for prediction of mortality (Fig. 1). We also excluded any irrecoverable or incomplete data. Cases related to death due to burns (ICD-10, T30.2; n = 1) and suicide attempts (T14.1, n = 1), with numbers counted in the subset and not the parent cohort, were excluded. No cases related to

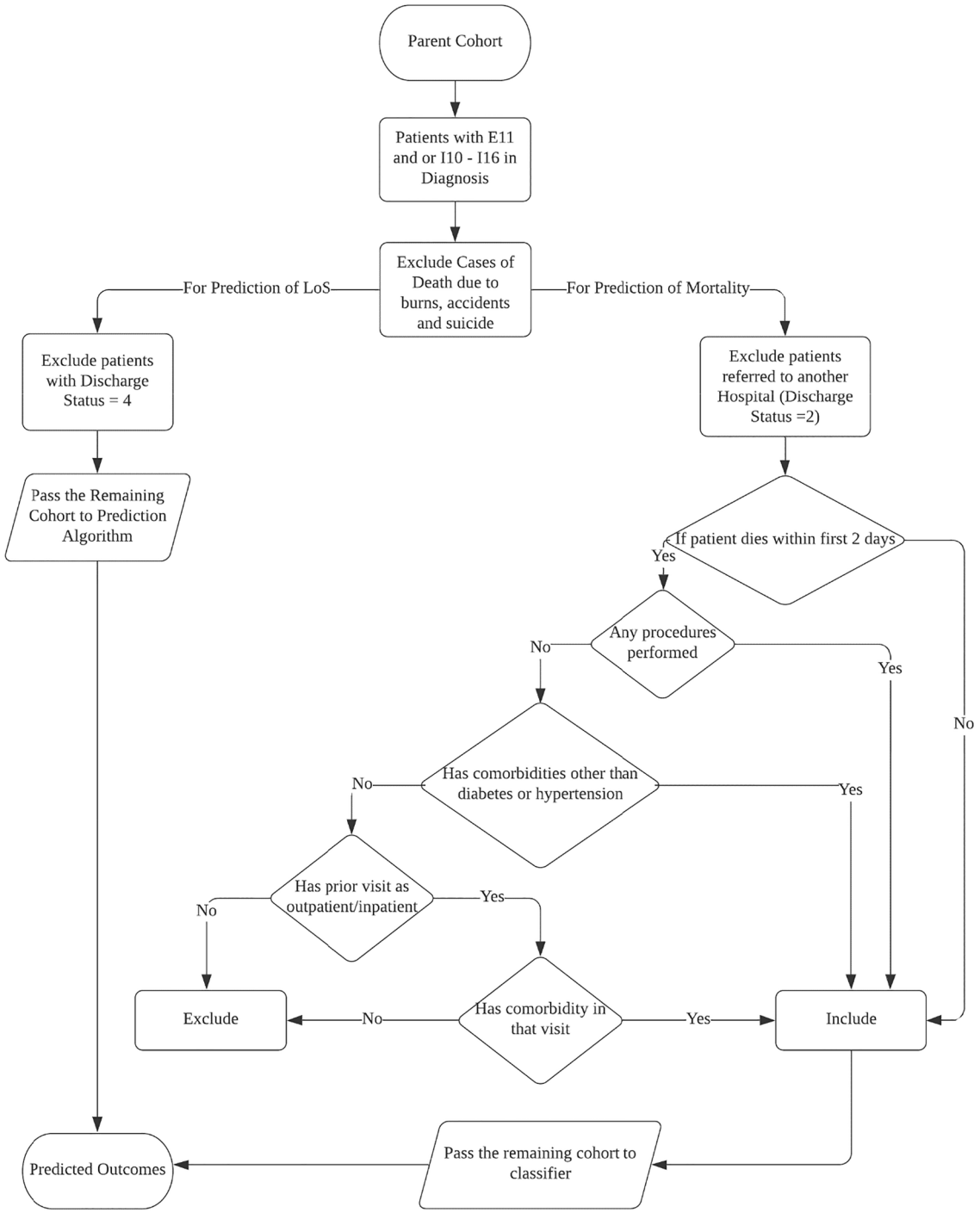


Fig. 1. Study design and cohort selection.

road accidents (including ICD codes) were found in the subset. Two deaths related to injuries, such as head trauma were excluded. Patients with no co-morbidities, those who died on the day of admission, and those with no prior record of admission to the hospital were also excluded (n = 3). All eligible participants were included in the analysis.

2.3. Features and outcomes

The outcome predicted using the model for mortality is a categorical variable with the values "alive" or "death", and LoS is predicted as a float-type variable. All the features were based on data obtained prior to discharge. Diagnosis codes identified in more than 5 patients were binarized (n = 216). Procedure codes identified in more than 5 patients were binarized (n = 35). The INA-CBGs code was divided into 4 columns: C1, C2, C3, and C4. C2 and C4 were left unaltered, while all features from C1 and C3 were binarized. C1 contained an alphabetic symbol, denoting a particular case-based group and C3 contained an exact number, denoting the disease from the case-based group. Age was categorized as young (0–35 years), middle-aged (36–58 years), old (59–80), and elderly (≥81). From a description of causes of death, ICD codes for the cause of death were used in the calculation of risk groups and patients with co-morbidities listed among the causes of death with a severity level of 3 were classified into high-risk groups. Patients with no high-risk co-morbidities listed but with a severity level of 3 were classified into high- and medium-risk groups. Patients who had co-morbidities listed among the high-risk co-morbidities but with a severity level of 1 or 2 were classified into low- and medium-risk groups. Patients who had neither high-risk co-morbidities listed nor a severity level of 3 were classified into low-risk groups. The cost of hospitalization was used only for analysis and was not used as a feature.

2.4. Development of predictive model

In total, we divided 80% of the data for training and validation and 20% for testing. Training data were further divided as follows: 80% for training and 20% for 10-fold cross validation. Table 1 shows that there are 1977 T2DM patients, 2903 HTN patients, 504 patients with both T2DM and HTN.

Non-normally distributed numeric variables were Box Cox transformed for modeling procedures. The outcomes for both mortality and LoS were predicted based on each patient visit using the current predictor values. In total, 303 features were obtained and then scored using different scoring methods such as Information Gain and ReliefF algorithm. Different numbers of features were selected on the basis of the score of each scoring method as well as on the basis of how the number affected the result. The best scored features were then entered into the predictive model.

2.5. Inpatient length of stay (LoS)

To predict inpatient LoS, the numeric target variable LoS was first predicted to be a longer duration of hospital stay or an average duration of hospital stay. This was then used as a feature to predict the numeric inpatient LoS. The models used to predict inpatient LoS were as follows:

I. A linear regression model, with elastic net regression, an alpha of 0.03, and elastic net mixing (L1:L2 = 0.83:0.17. 2).
II. An SVM model, with SVM type v-SVM regression cost (C) and complexity bound (v) being 1 and 0.50, respectively. The kernel used was polynomial $(g\ x.y + c)^d$ with a numerical tolerance of 0.0025 and maximum iterations (250).
III. A stochastic gradient descent model, with a classification loss function as Huber, a regression loss function as e sensitive (with e: 0.01), regularization using elastic net, and a constant learning rate (with 1200 iterations).
IV. A random forest model, comprising the number of trees (n = 34), the number of attributes considered at each split (n = 14), and growth control of not splitting subsets (<20).

Table 1
Study population.

Total Number of Patients			Diabetic	Hypertension	Both
4376	Male	1974	1977	2903	504
	Female	2402			

2.6. Mortality

The model for prediction of mortality used a Machine Learning for Networking (MLN) network with back propagation and neurons per hidden layer (230 defined as the ith element represents the number of neurons in the ith hidden layer). The number of maximum iterations (310, ReLu) was used as activation function for the hidden layers. The Alpha L2 penalty (regularization term) parameter was 0.0005, and a stochastic gradient descent was used as a solver for weight optimization. All other parameters were sklearn defaults. Logistic regression, random forest, and support vector machine (SVM) models were also used to predict mortality.

The ratio of dead to alive patients was 1:10. The cohort of dead patients was up sampled to 1:5 to address data imbalance. The ratio was changed only in the training data set.

3. Results

3.1. Study population characteristics

The mean age of the inpatients with T2DM was 57.0 years (standard deviation [SD], 12.7; men, 42.29%; women, 57.71 %). The mean age of the inpatients with HTN was 58.4 years (SD, 13.6; men, 45.92%; women, 54.08 %). There was a higher proportion of patients with HTN (66.34%) than patients with T2DM (45.18%). The median (interquartile range) inpatient LoS and cost of hospitalization for patients with T2DM were higher (6 days (4–9); mean cost, 4,976,800 Indonesian rupiah [IR]). Median cost 4,733,400 IR (385–1553)) than those for patients with HTN (5 days; range, 1–38); mean cost, 4,275,900 IR (95% CI 825–1003), and; median cost, 3,925,200 IR (range, 494–1147). We found that patients with T2DM had a higher mortality rate (10.27%) than those with HTN (5.82%). Patient characteristics are shown in Table 2.

The frequency distribution of inpatient LoS is shown in Fig. 2 and indicated that the maximum LoS was 6 days. Fig. 3 shows the number of patients who died and those who were discharged, as classified according to the respective physicians. The probability of patients being discharged was because of their own request or referral to a hospital with a higher level. The number of admissions, classified for each physician, for the specified LoS is shown in Fig. 4, where most of the admissions had LoS<12.1 days.

3.2. Prediction of inpatient length of stay (LoS)

The linear regression model performed best in terms of predicting the LoS with an R2 of 0.53 (RMSE, 2.43; mean absolute error [MAE] 1.722). The Gradient Boosting Regression model also performed to a similar level (R2, 0.52; RMSE, 1.97; MAE, 1.103). Stochastic gradient descent with a regression loss function, Huber, and random forest were found to be average performing models. **ReliefF** was used as a scoring method with 73 best ranked features. The evaluation metrics of the best performing models are shown in Table 3.

Socio-demographic variables, diagnosis variables, and other features independently and highly correlated to LoS were: the INA-CBGs code (β = 0.773), which is a primary diagnosis; physician-in-charge (β = 0.711); age (β = 706), and; severity level (β = 0.533).

Table 2
Characteristics of the patients, reported as n(%);

Characteristics	T2DM (n = 1977)	HTN (n = 2903)	Total Patients included in the study (n = 4376)
Sex			
Male	836(42.29)	1333(45.92)	1974(45.11)
Female	1141(57.71)	1570(54.08)	2402(54.89)
Age (mean ± SD)	57.0 ± 12.7	58.4 ± 13.6	57.9 ± 13.5
<35	70(3.54)	149(5.13)	205(4.68)
35–57	945(47.80)	1169(40.27)	1879(42.94)
58–80	930(47.04)	1486(51.19)	2166(49.50)
greater than80	32(1.62)	99(3.41)	126(2.88)
Discharge status			
Physician's permission	1756(88.82)	2706(93.21)	3985(91.06)
Referral	16(0.81)	24(0.83)	36(0.82)
Patient request	2(0.10)	4(0.14)	6(0.14)
Death(In-hospital mortality)	203(10.27)	169(5.82)	349(7.98)
Room Type			
Class 1(2–4 beds)	549(27.77)	622(21.43)	1029(23.51)
Class 2(3–5 beds)	299(15.12)	454(15.64)	668(15.27)
Class 3(4–6 beds)	1129(57.11)	1827(62.93)	2679(61.22)
Risk group			
High risk	158(7.99)	244(8.41)	369(8.43)
High-medium risk	5(0.25)	17(0.59)	22(0.50)
Low-medium risk	1614(81.64)	2238(77.09)	3427(78.31)
Low	200(10.12)	404(13.92)	558(12.75)
No. of co morbidities			
0 co morbidities	152(7.69)	209(7.20)	361(8.25)
1 co morbidity	674(34.09)	1090(37.55)	1624(37.11)
2 co morbidities	654(33.08)	971(33.45)	1424(32.54)
3 to 5 co morbidities	473(23.93)	608(20.94)	927(21.18)
5 + co morbidities	24(1.21)	25(0.86)	40(0.91)
Hospital Cost			
Average Cost(min–max)	4976800(1410500–34804500)	4275900(141500–37037300)	4560200(141500–37037300)
Median(IQR)	4733400(3400700–5920300)	3925200(2840700–4867100)	4024000(3150900–5495300)
Length of stay			
Average LOS(min–max)	7.3(1–34)	6.2(1–38)	6.6(1–38)
Median(IQR)	6.0(4–9)	5.0(4–8)	6.0(4–8)
Severity Level			
Level 1(less serious)	1058(53.52)	1596(54.98)	2381(54.41)
Level 2(medium serious)	756(38.24)	1046(36.03)	1604(36.65)
Level 3(very serious)	163(8.24)	261(8.99)	391(8.94)
Type of patient			
Procedure Inpatient	111(5.61)	166(5.72)	259(5.92)
non-procedure Inpatient	1863(94.23)	2730(94.04)	4107(93.85)
Midwifery Inpatient	3(0.15)	7(0.24)	10(0.23)

IQR: inter quartile range; The Hospitalization Cost in Indonesian Rupiah.

A list of ReliefF scores concerning 45 highly correlated features is shown in Table 4. We noted that the features- hospital room type and the risk group, which were created to predict mortality, played a role in predicting LoS along with common and critical co-morbidities, such as diseases of the circulatory system, anemias, and metabolic disorders. The number of patients according to thirty most common comorbidities for T2DM and HTN in our dataset are shown in Table 5 and Fig. 5 shows the actual and predicted inpatient LoS.

3.3. Prediction of mortality

The results were validated using a K-fold cross-validation and the random sampling technique. For both techniques, the validation sample was 20% of the training set (80% of the entire data set). The best results were obtained using MLP with back propagation, with an AUROC of 0.899 and a PPV and NPV of 69.33 and 98.61, respectively (Table 6 and Fig. 6). The random forest classifier performed similarly to the MLP (AUROC, 0.872; PPV, 85.33; NPV, 89.79) with the information gain feature selection selecting 73 best ranked features. The scores for the best 15 are provided in Table 4. Other prediction models with their evaluation scores are listed in Table 7.

Socio-demographic variables and other features found to be independently and highly correlated to in-hospital mortality were as follows: diseases of the circulatory system ($\alpha = 0.069$), primary diagnosis ($\alpha = 0.066$), age ($\alpha = 0.048$), physician-in-charge ($\alpha = 0.028$), and risk group ($\alpha = 0.023$). It is noteworthy that the feature physician on duty correlated more closely with in-patient mortality than risk group, and sex was not among the 15 highest-ranked features. Brain disorders and sepsis were the second and third most common co-morbidities responsible for death after diseases of the circulatory system.

4. Discussion

This study employed neural network techniques to predict LoS and mortality risk for inpatients in a tertiary hospital setting in Indonesia with any of the two primary diagnoses: a primary discharge diagnosis of T2DM (E11) or HTN (I10-I15). To the best of our knowledge this is the first study to present data from Indonesia for prediction of LoS and mortality risk among T2DM and HTN patients. No previous study on Indonesian data has conducted similar prediction modelling. Our findings showed a high correlation between the attending physician and the prediction of inpatient LoS and mortality risk. Tables 3 and 5 show that this correlation was listed higher than other findings and was more closely correlated (Figs. 3 and 4), with LoS and cases of mortality listed higher for some physicians than others. This feature provided an additional predictive ability for inpatient LoS and mortality. The inpatient LoS varied from 1 to 38 days, with a significant number of cases of a duration more than 6 days.

While predicting a longer LoS, most studies have designed prediction models to determine whether an admission might involve a longer LoS

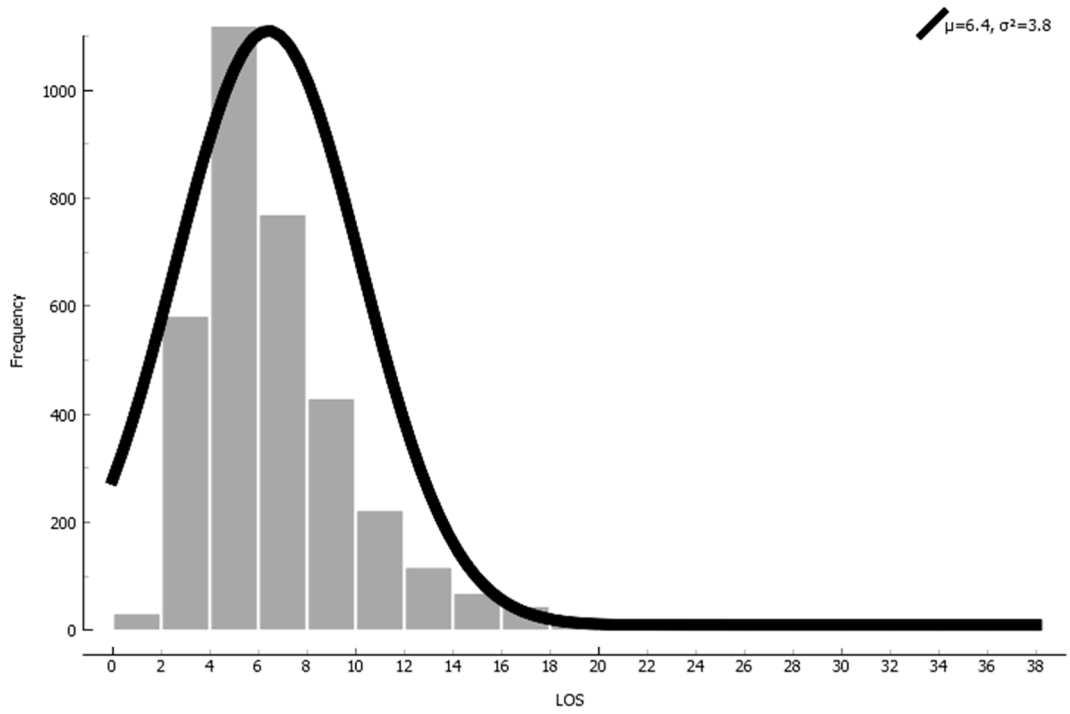


Fig. 2. Inpatient length of stay frequency distribution.

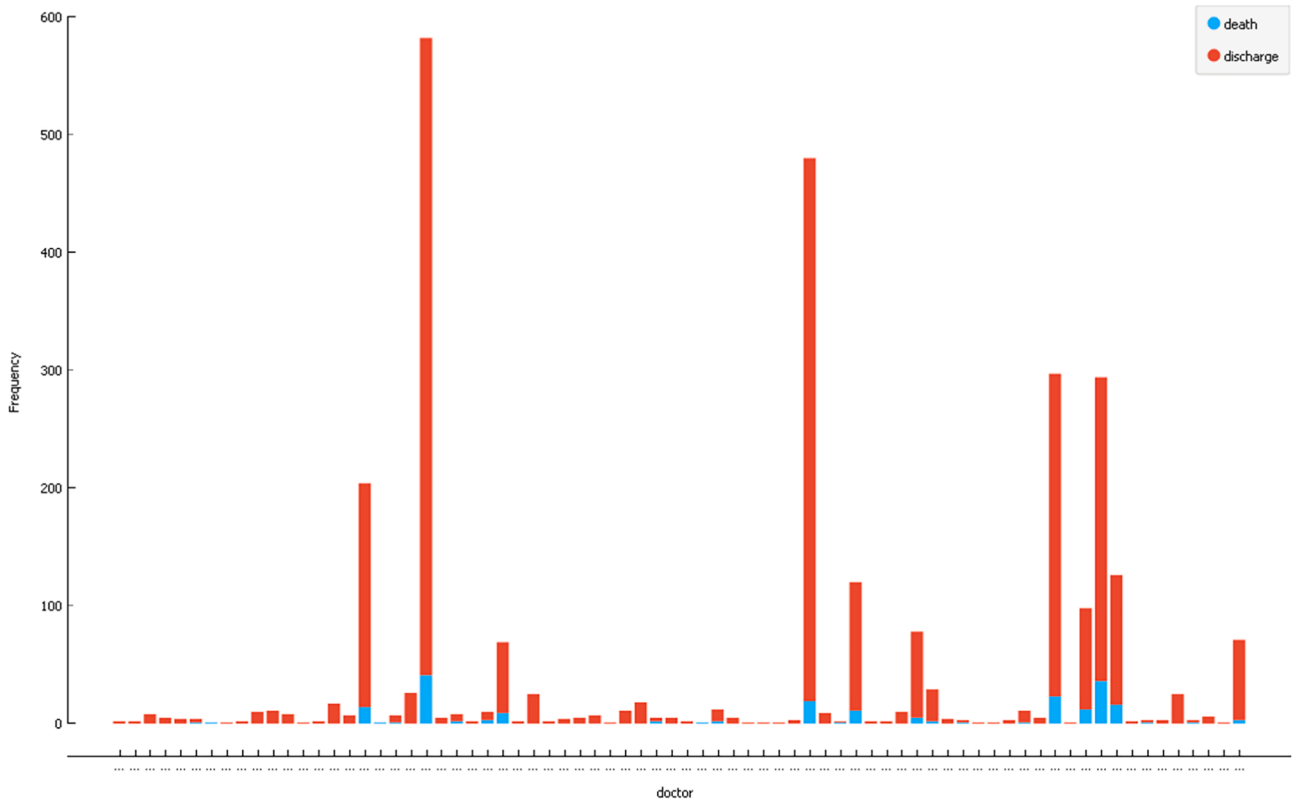


Fig. 3. Number of patients who died or discharged during an encounter.

[39–41] or whether the admission might be within a fixed LoS range [42,43]. In our study, we determined what factors can be correlated and used to predict the LoS and mortality.

In this study, the mean LoS in the training dataset and in the test dataset was 6.2 days and 6.4 days, respectively, for patients with T2DM and HTN. With 73 input variables, our models could correctly predict inpatient LoS up to 9 days, as shown in Fig. 6. In comparison, the linear regression model could predict inpatient LoS greater than 14 days, as shown in Fig. 6.

We observed that the linear regression model generally performed

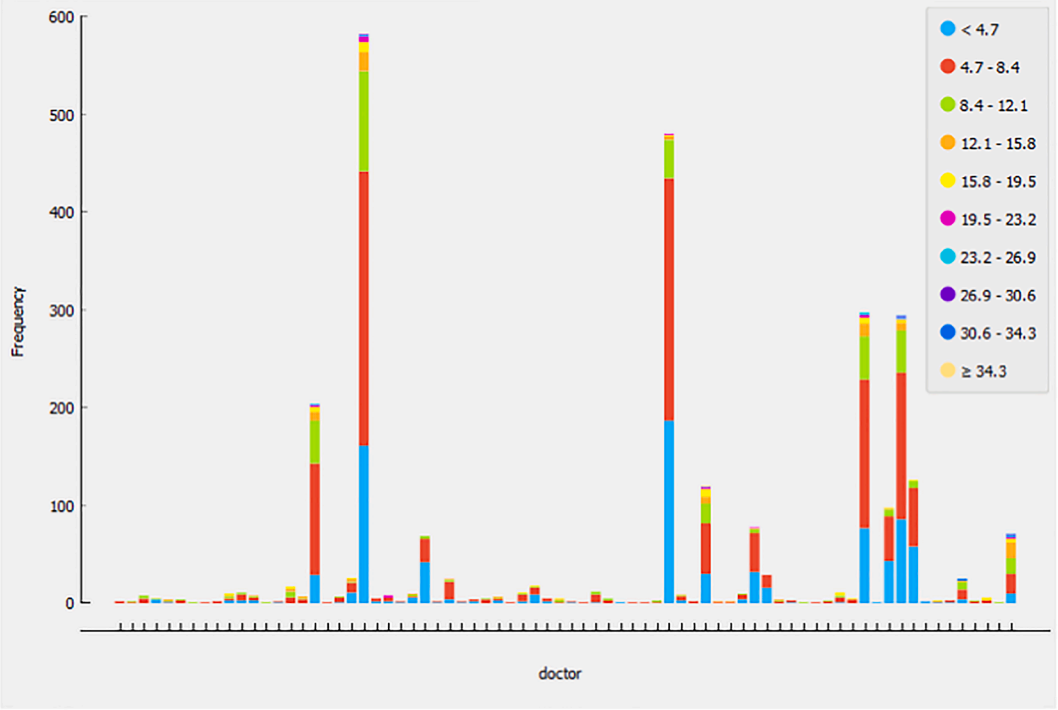


Fig. 4. Number of patient admissions and discretized LoS.

Table 3
LoS prediction performance of various models.

Model	MSE	RMSE	MAE	R2
Linear Regression	5.95	2.43	1.722	0.53
Gradient boosting Regressor	3.911	1.97	1.103	0.69
Random Forest	3.844	1.96	0.935	0.70
Stochastic Gradient Descent	6.08	2.46	1.736	0.52

Table 4
Best ranked features by ReliefF feature selection.

Index	Feature name	ReliefF score(β)
1.	INA-CBGs code based on primary diagnosis	0.773
2.	physician in-charge	0.711
3.	Age	0.706
4.	Severity level	0.533
5.	Diabetes	0.500
6.	Hypertension	0.403
7.	Sex	0.390
8.	Diseases of Circulatory System	0.386
9.	Hospital room Type	0.336
10.	Aplastic and other Anemias	0.201
11.	No. of co-morbidities	0.200
12.	Other chronic obstructive pulmonary Disease	0.195
13.	Endocrine nutritional and metabolic disease	0.141
14.	Risk group	0.137
15.	Ischemic heart Disease	0.109

slightly better than the SVM models in terms of MAE and R2, as shown in Table 4. This may be because the linear regression model was built using only 80% of the training dataset, and was comprised of data from the first 3 years, and the test dataset, which comprised the remaining 10 months of data, had been highly consistent with the previous 3 years. This implied that the clinical pathways were well established in our case hospital.

Comorbidities were the top fifteen features to influence LoS. The five most common co-morbidities observed were ischemic heart disease (n = 1026 patients), metabolic disorders (n = 595 patients), aplastic and other anemias (n = 495 patients), functional dyspepsia (n = 430 patients), and renal failure (n = 337 patients). Previous studies did not investigate the influence of comorbidities on LoS, while they focused on the order of prevalence of comorbidities. One study examined patients with T2DM and reported that the highest co-prevalence was

Table 5
Thirty most common co-morbidities.

Index	Disease name	Patient count
1	Ischemic heart disease	1026
2	Metabolic disorders	595
3	Aplastic and other anemias	495
4	Functional dyspepsia	430
5	Renal failure	337
6	Other chronic obstructive pulmonary disease	272
7	Cerebral infarction	239
8	Atrial fibrillation and flutter	218
9	General symptoms and signs	167
10	Intestine infectious disease	160
11	Medical observation	156
12	Glucose regulation and pancreatic disorders	124
13	Other diseases of digestive system	119
14	Heart failure	117
15	Gastro-oesophageal reflux disease	99
16	Pneumonia	98
17	Dorsopathies	91
18	Symptoms and signs involving cognition, perception, emotional state and behavior	84
19	Other cardiac arrhythmias	82
20	Disorders of gall bladder biliary tract and pancreas	80
21	Infections of skin and subcutaneous tissue	79
22	Eye disorders and ear	74
23	Hemorrhage	70
24	Other disorders of kidney, ureter and urinary system	70
25	sepsis	69
26	Other sequelae of other cerebrovascular disease	67
27	Other heart disease	66
28	Diseases of liver	66
29	stroke	64
30	Disorders of female genitalia	63

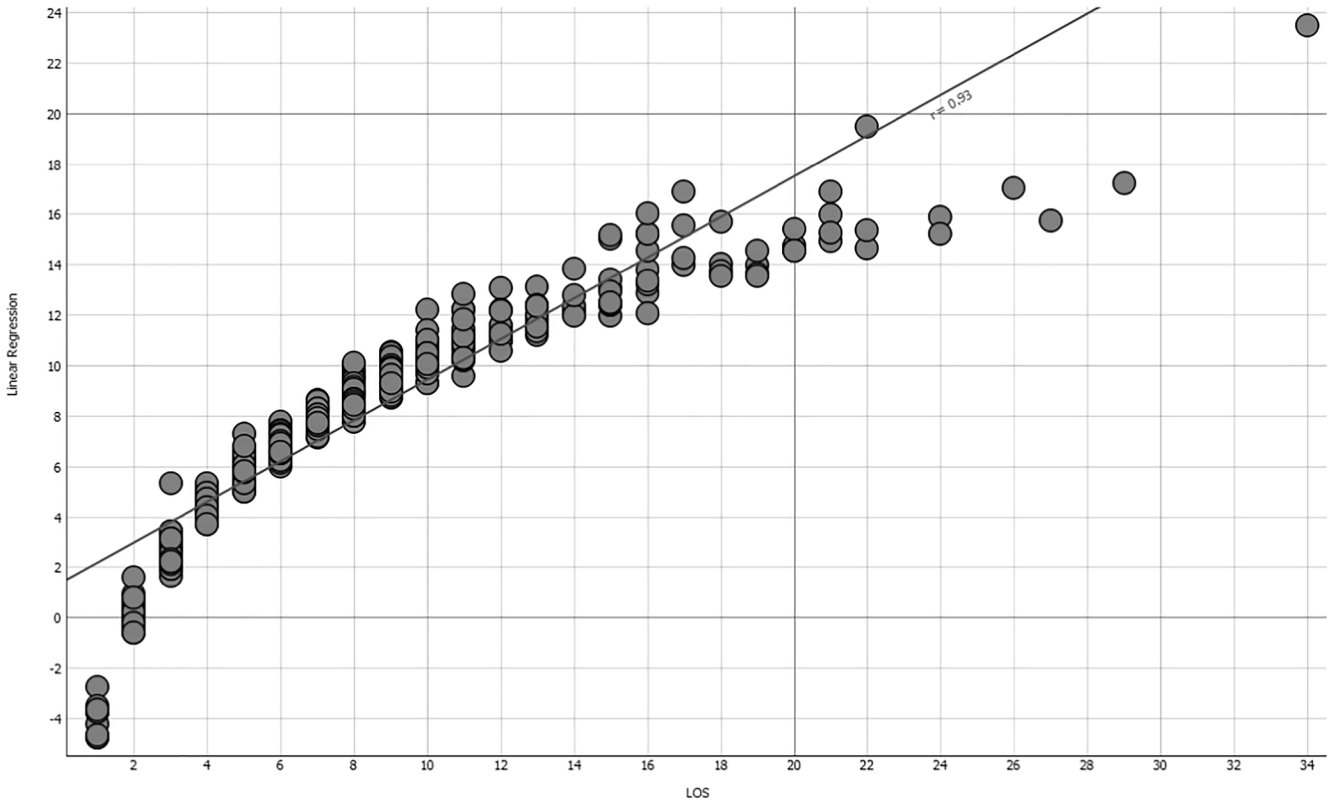


Fig. 5. Predicted inpatient LoS using linear regression.

Table 6
Mortality prediction performance of various models.

	Multilayer Perceptron	Logistic Regression	Random Forest	Support Vector Machine
AUROC	0.899	0.834	0.872	0.887
CA (Classification Accuracy)	0.910	0.851	0.898	0.904
PPV (%)	69.33	82.67	85.33	86.66
NPV (%)	98.61	85.62	89.79	90.79
F1	0.978	0.914	0.942	0.945
Recall	0.986	0.854	0.904	0.907

demonstrated in those with HTN and hyperlipidemia (67.5%), followed by patients with HTN who were overweight/obese (66.0%), overweight/obese patients with hyperlipidemia (62.5%), patients with HTN and chronic kidney disease (CKD) (22.4%), and patients with hyperlipidemia and CKD (21.1%), [44]. Our study showed unique findings ~~found~~ that the most influencing comorbidity affecting the LoS for T2DM and HTN is ischemic heart disease. In another study significant differences in terms of the average LoS between patients with T2DM and HTN and those without HTN ($P < 0.001$) and between patients with T2DM and neuropathy and those without neuropathy ($P < 0.001$) [45]. Similar to our findings, most inpatients with T2DM and with a hospital LoS greater than 5 days were mostly females (57.7 %) and aged 45–65 years [46,47] . The primary disease identified among the comorbidities in our study was heart failure (n = 117 inpatients), which was similar to findings reported by Chen et al [48]. The study showed that participants with HTN at the time of diagnosis of T2DM showed higher rates of all causes of death and cardiovascular events compared with normotensive patients with T2DM, which is also consistent with our findings. As observed in our findings, Salim et al also reported significant differences in the average LoS between patients with T2DM who have coronary heart disease and those without coronary heart disease ($P = 0.035$) [45].

Of the patients with HTN and T2DM, those aged 58–80 years comprised the largest age group (n = 2166 ± 49.50). This finding is similar to Kabakov et al.'s findings that showed that the prevalence of HTN clearly increases with age [49]. Most patients with T2DM have been reported to range in age from 56 to 63 years (27.86%) [4647]. Our study showed that mortality was higher in T2DM (10.27%) than HTN (5.82%). Females were more prone towards T2DM (57.71%) than HTN (54.08%). Patients aged 35 – 80 years old suffering from HTN were higher than T2DM, however, the mortality rate was higher among patients with T2DM.

The four types of discharge status involved a physician's permission for patient discharge, a patient referral, a patient request for discharge, and death (in-hospital mortality). Most discharges involved a physician's permission (n = 3985; 91.06 %) and inpatient deaths (n = 349, 7.98%). Physicians are listed as the highest-ranked feature according to the information gained from feature selection for the prediction of mortality. According to our study result, the late admission of several inpatients has resulted in their deaths 1 or 2 days later after arriving in the hospital. Another possibility of death could be when patients fail to attend polyclinics in hospital to undertake treatment, and physicians do not have an opportunity to provide relevant health-related information. This influence of the physician's role on the LoS and mortality is another unique finding of our study. Previous research demonstrated the influence of demographic factors, arrival time etc on the LoS of patients [50]. Our study highlights the physician's role in educating the patients to seek early treatment, as patients often attend a hospital only when their condition is severe. The physician is responsible not only for providing treatment recommendations but also taking into consideration the high risk patients as well as those who can benefit from therapy [51]. Thus, the physician plays an important role in the assessment of the different kind of problems of the patient and the probable risk factors. They need to be alert that due to individual risk factors, different patients may react differently in terms of treatment outcomes.

This study had some limitations. First, some data were unavailable

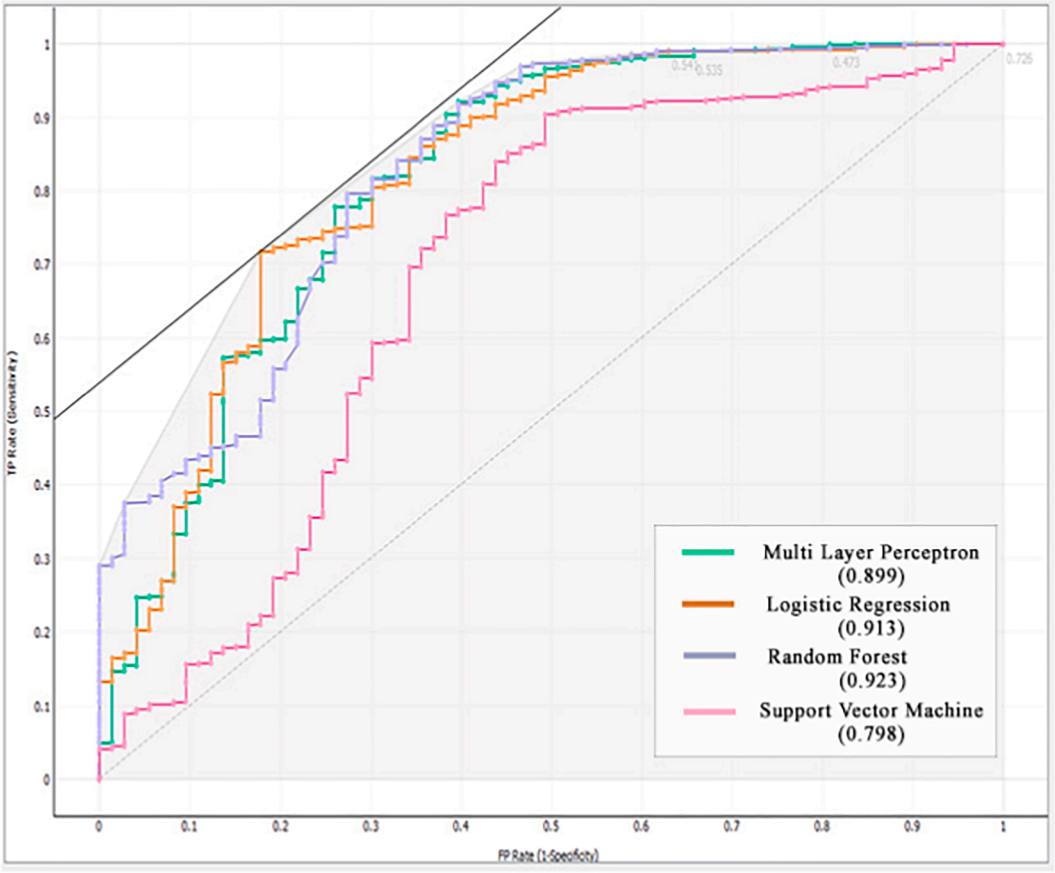


Fig. 6. Receiver operating characteristic curve for all models.

Table 7
Best ranked features by information gain feature selection.

Index	Feature name	Information gain score (α)
1.	Diseases of Circulatory System	0.069
2.	INA-CBG CODE based on primary diagnosis	0.066
3.	Age	0.048
4.	physician in-charge	0.028
5.	risk group	0.027
6.	brain disorders	0.023
7.	sepsis	0.022
8.	Severity level	0.020
9.	hypertensive	0.019
10.	hemorrhage	0.016
11.	other respiratory failures and disorders	0.015
12.	No. of co-morbidities	0.012
13.	Cardiac arrest	0.013
14.	T2DM	0.013
15.	Metabolic disorders	0.010
16.	Functional dyspepsia	0.009

concerning critical patients who had been discharged alive, those who had not been transferred to any other acute care hospital, and those who had no record of readmission after the discharge date. Therefore, we predicted mortality risk during the inpatient period. We assumed that, for patients with a discharge status of 4 (indicating patient death), the patient had died while in hospital care.

Our study provides methods for individual prediction of hospital LoS and mortality based on easily available demographic and clinical data. These predictions can improve the hospital quality management in handling the patient, help clinician to take decision for the next patient's care, improve the patient's safety. Our study provides scope for development of individual patient strategies for better outcomes, while aiding in clinical decision support, personalized treatment and ensuring patient-centered safe and effective care. Further, external evaluation of these prediction models is indicated to establish validation of our methodology. Future research is required for more detailed description of the hospital LoS, details of discharge of the patient and reason of mortality.

5. Conclusion

We examined several algorithms to predict inpatient LoS and mortality. The MLP best predicted mortality, and the linear regression model best predicted inpatient LoS. T2DM and/or HTN-related co-morbidities could be the possible cause for an extended inpatient LoS. The physician's role has a significant effect on patient outcomes in relation to how frequently healthcare information is provided to patients. Our model gives the good prediction of LoS and mortality, and it can be used to manage the patient safety in the hospital and hospital management.

6. Summary points

What was already known on the topic

• Type 2 diabetes mellitus (T2DM) and hypertension (HTN) places a considerable economic burden on medical service frameworks.

• Real world claims-based INA-CBGs dataset can facilitate the development of artificial Intelligence algorithms for predicting LoS and mortality.

• Predicting LoS and mortality among hospitalized patients with T2DM and HTN using real world claims data is poorly investigated.

What this study added to our knowledge

• Development of Artificial intelligence algorithms to predict LoS and mortality based among diabetic and hypertensive patients.

• Our finding showed a high correlation between the attending physician and the prediction of inpatient LoS and mortality risk.

• The Linear Regression and The Gradient Boosting Regression model performed best in terms of predicting the LoS. MLP model performed in predicting inpatient mortality.

7. Ethics approval

This study was approved (No. 044/EC-KEPK/IX/2020) by the Health Research Ethics Committee of Polytechnic of Health Ministry Tasikmalaya, Tasikmalaya, West Java, Indonesia.

Author contributions

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Availability of Data and Materials

The data that support the findings of this study are not publicly available and requires ethics approval from RSUD Dr. Soekardjo hospital. However, aggregated data are available from authors upon reasonable request and with permission of RSUD Dr. Soekardjo hospital.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmedinf.2021.104569.

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